

Microservices Real-world Use Cases

The top real-world examples of microservices adoption are:

- Amazon — over 1000 microservices
- Netflix — over 700 microservices
- Uber — over 500 microservices



Microservices Real-world Use Cases . . . Amazon

- Like many startups, Amazon was built as a monolith because of its ease.
- Amazon started growing rapidly, so it needed to develop many new features and scale its online platform fast.
- Amazon's codebase became extremely large and complicated in 2001. It had to implement hundreds of changes.
- **Problem:** Developers had to spend lot of time resolving conflicts & merging all changes into a single version, leading to an unacceptable product delivery timeline.
- Developers were required to rebuild the codebase and run a lot of tests to ensure that a new version of the product worked correctly.
- Striving to simplify the pipeline, the company's software engineers developed a revolutionary idea. They decided to decompose the monolith.
- They thoroughly analyzed codebase and allocated functional units that served the only purpose. Developers pulled the items out to create independent microservices and connected them using API integrations.
- Amazon engineers analyzed all product development processes & started developing automation solutions to eliminate idle time in their CI/CD pipeline.

Microservices Real-world Use Cases . . . Netflix

- Netflix is the youngest business case for microservices adoption.
- First web app launched by the company was a monolith.
- With growing popularity, the company faced a lot of challenges with processing all the requests submitted by its customers leading to many database failures.
- Netflix decided to replace its on-premises database with a cloud-based solution from AWS.
- Cloud servers can be easily scaled horizontally to get the required amount of resources required to process all the requests.
- Developers were required to decouple the monolith architecture to move the database to the cloud.
- After noticing the positive impact of using dissociated architecture, the company spent years switching from a monolithic architecture to microservices.

Microservices Real-world Use Cases . . . Uber

- Uber was launched as a monolith with one codebase that handles all the processes.
- Monolithic architecture implied all required functionality to connect drivers with passengers, process payments, manage trips, etc.
- With rapid growth, Uber faced lot of challenges with scaling its services faster.
- Monolithic architecture needed developers to redeploy the codebase to implement new changes.
- Even minor changes affected the entire codebases, so it was necessary to conduct comprehensive testing to release new features.
- Being unsatisfied with number of developer resources and time every new feature consumed, Uber decided to break its application into several microservices.
- Microservices helped Uber scale up different services like **billing, trip management, driver management, and notifications independently.**
 - Developers don't have to examine / update entire codebase to release minor updates.